Q] A person allows a 10% discount for cash payment from the marked price of a toy and still he makes a 10% gain. What is the cost price of the toy which is marked Rs 770?

① $\% D = \dfrac{MP - SP}{MP} \times 100$

$SP = \overset{770}{\cancel{MP}} \left[1 - \dfrac{\overset{10}{\cancel{\% D}}}{100} \right]$

$\Rightarrow SP = 770 \left[\dfrac{9}{10} \right] = 693$ //

② $\% P = \frac{SP - CP}{CP} \times 100$

$$\overset{693}{\cancel{SP}} = CP\left[1 + \frac{\overset{10}{\cancel{\% P}}}{100}\right]$$

$$693 = CP\left[\frac{11}{10}\right]$$

$$\boxed{CP = ₹\,630}$$

Q] A trader fixed the price of an article in such a way that by giving a rebate of 10% on the price fixed, he made a profit of 15%. If the cost of the article is Rs 72 the price fixed on it is

① $SP = \underline{\underline{MP}}\left[1 - \frac{D\%}{100}\right]$ 10

② $SP = CP\left[1 + \frac{\%P}{100}\right]$ 72, 15

$\Rightarrow 72\left[1+\frac{15}{100}\right] = MP\left[1-\frac{10}{100}\right]$

$\boxed{MP = ₹92}$

Q] A man purchases two clocks A and B at a total cost of Rs 650. He sells A with 20% profit and B at a loss of 25% and gets the same selling price for both the clocks. What are the purchasing prices of A and B respectively?

① $SP_A = CP_A\left[1 + \frac{20}{100}\right]$

② $SP_B = CP_B\left[1 - \frac{25}{100}\right]$

③ $SP_A = SP_B$

$$CP_A\left[1+\frac{\cancelto{1}{20}}{\cancelto{5}{100}}\right] = CP_B\left[1-\frac{\cancelto{1}{25}}{\cancelto{4}{100}}\right]$$

$$CP_A\left[\frac{6}{5}\right] = CP_B\left[\frac{3}{4}\right]$$

M1 $$\frac{CP_A}{CP_B} = \frac{3/4}{6/5} = \frac{\cancel{3}\times 5}{{}_2\cancel{6}\times 4} = \frac{5k}{8k}$$

$$\Rightarrow \quad CP_A = 5k$$

$$CP_B = 8k$$

$$8k + 5k = 650$$

$$k = \frac{650}{13}$$

$$\therefore CP_A = 5k = 5 \times \frac{650}{13} = 250$$

$$CP_B = 8k = 8 \times \frac{650}{13} = 400$$

M2

$$\frac{CP_A}{CP_B} = \frac{3/4}{6/5} = \frac{\cancel{3}\times 5}{{}_2\cancel{6}\times 4} = \frac{5}{8}$$

Total $CP = 650$

$$\Rightarrow \quad CP_A = \frac{5}{5+8} \times 650 = 250 \,//$$

$$\Rightarrow \quad CP_B = \frac{8}{5+8} \times 650 = 400 \,//$$

Q] A trader loses 20% by selling an article for Rs 480. If he is to gain 20% , he should sell it for rupees ?

① $\overset{480}{\cancel{SP}} = CP\left[1 - \frac{20}{100}\right] \Rightarrow CP = 600$

② $SP = \overset{600}{\cancel{CP}}\left[1 + \frac{\overset{1}{\cancel{20}}}{\underset{5}{\cancel{100}}}\right] = 720$ //

Q] A grocer bought 10 kg of apples for Rs 81 out of which one kg was found rotten. If he wishes to make a profit of 10% at what rate per kg he should sell it?

1 $10\text{ kg} \rightarrow ₹81$

-1 kg [rotten]

2 $CP = 9\text{ kg} \rightarrow ₹81 \Rightarrow \boxed{1\text{ kg} \rightarrow ₹9}$

3 $SP = \overset{81}{\cancel{CP}}\left[1 + \frac{10}{100}\right] = 89.1 \ (9\text{ kg})$

SP $\rightarrow$ 1kg $\rightarrow$ 9.9 ₹

Q] A man sold an article for Rs 6800 and incurred loss. Had he sold the article for Rs 7850 his gain would have been equal to half the amount of loss that he incurred. At what price should he sell the article to have 20% profit?

1 Loss :

$SP = 6800$

$CP = x$

$Loss = CP - SP$

$$\boxed{Loss = x - 6800}$$

(2) $SP = 7850$

$CP = x$

$\text{Gain} = P = SP - CP = \underline{7850 - x}$

(3) $\text{Gain} = \frac{1}{2} \text{Loss}$

$$(7850 - x) = \frac{1}{2}(x - 6800)$$

$$\Rightarrow x = 7500$$

(4) $SP = CP\left[1 + \frac{\cancelto{1}{20}}{\cancelto{5}{100}}\right]$

$= 7500\left[\frac{6}{5}\right]$

$SP = 9000$ //

Q] A person divided an amount Rs 1,00,000 into two parts and invested in two different schemes. In one he got 10% profit and in the other he got 12%. If the profit percentages are interchanged with these investments he would have got Rs 120 less. Find the ratio between his investments in the two schemes.

(A) 11:14
(B) 47:53
(C) 37:63
(D) 9:16

[GATE 2019 ME]

1. 100K

S2: 12% | 10%

$\boxed{x}$ $\quad$ $\boxed{100K - x}$

S1: 10% | 12%

2. $10\% \text{ of } x + (100K - x) \times 12\% \rightarrow S1$

$12\% \text{ of } x + (100K - x) \times 10\% \rightarrow S2$

$$S_1 - S_2 = 120$$

10% of x + $(100k - x) \times 12\%$

$-$ $= 120$

12% of x + $(100k - x) \times 10\%$

$$\Rightarrow \left[\frac{10x}{100} + \frac{(100k - x) \cdot 12}{100}\right] - \left[\frac{12x}{100} + \frac{(100k - x)10}{100}\right] = 120$$

$$\Rightarrow \left[\frac{10 \cdot x}{100} + \frac{(100k - x) \cdot 12}{100}\right] - \left[\frac{12x}{100} + \frac{(100k - x)10}{100}\right] = 120$$

$$x = 47000$$

$$100k$$

$$\boxed{x} \qquad \boxed{100k - x}$$

$$47k \qquad 53k \qquad \Rightarrow \frac{47}{53}$$

Q] Round trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is Rs 100, a group of 5 tourists purchasing round trip tickets will be charged Rs 850 [GATE 2014 CS]

① 1 way : 1P = ₹100

2 way = Round Trip ⇒ 1P = ₹200

② 5P = Round Trip ⇒ 5P = 1000 ₹

· Discounts

1. 10% D on total fare

$$\frac{10}{100} \times 1000 = 100 ₹$$

2. 5% D on total fare

$$\frac{5}{100} \times 1000 = 50 ₹$$

SP = Round trip ⇒ SP = 1000 ₹

↓

Discounts = 150

↓

Total fare after discount = 1000 − 150

= 850 ₹ //

Q] The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of Rs 50 per unit. The number of units to be produced by the firm such that the profit is maximized will be ___5___

[GATE 2012 ME/CE]

① $CP = 5q^2$

$SP = \frac{50 ₹}{unit} \times q \text{ units} = 50q$

$P = SP - CP = 50q - 5q^2$

$$f(q) = 50q - 5q^2$$

$$\frac{df}{dq} = 0$$

$$\Rightarrow \boxed{50 - 10q} = 0$$

$$\Rightarrow q = \frac{50}{10} = 5$$ [max pnt]

$$\Rightarrow f'' = -10$$

$$f''(5) = -ve < 0$$ [max]

* Compounded ratio

$$\frac{a}{b}, \frac{c}{d}, \frac{e}{f}$$

$\downarrow$

$$\text{Compounded ratio} = \frac{ace}{bdf} \quad //$$

$\therefore \frac{A}{B} = \frac{p}{q}$

Duplicate ratio of $A : B = p^2/q^2$

Triplicate ratio of $A : B = p^3/q^3$

Sub-Duplicate ratio of $A : B = \sqrt{p}/\sqrt{q}$

Sub-triplicate ratio of $A : B = \sqrt[3]{p}/\sqrt[3]{q}$

8] $\frac{A}{B} = \frac{4}{9}$

what is triplicate ratio of [Sub-duplicate ratio of A & B].

Sol^n: ① Sub-duplicate ratio of A : B = $2/3$

② Triplicate of $\frac{2}{3} = \frac{2^3}{3^3} = \frac{8}{27}$ //

Q] Present ages of 3 brothers are in ratio 3 : 4 : 5 . After 10 yrs sum of their ages will be 78 . Find their ages now ?

[ESE] 2017

Sol^n :

① Present ages : $3x, 4x, 5x = 12, 16, 20$ //

② $(3x+10)+(4x+10)+(5x+10)$

$= 78$

$x = 4$